

EXERCISE 3.3:

Q1:

Let $x(u, v) = (u, v, auv)$ be a parametrization of the hyperboloid. Then

$$x_u = (1, 0, av), \quad x_v = (0, 1, au), \quad x_{uv} = (0, 0, a), \quad x_{vu} = (0, 0, a)$$

$$x_{uu} = x_{vv} = (0, 0, 0) \quad \text{Therefore } E = \langle x_u, x_u \rangle = 1 + a^2 v^2$$

$$F = \langle x_u, x_v \rangle = a^2 uv$$

$$G = \langle x_v, x_v \rangle = a^2 u^2 + 1$$

$$\Rightarrow E(0,0) = 1 = G(0,0), \quad F(0,0) = 0$$

$$N(u, v) = \frac{x_u \wedge x_v}{|x_u \wedge x_v|} = \frac{1}{\sqrt{1+a^2(u^2+v^2)}} (-av, -au, 1) \Rightarrow N(0,0) = (0,0,1)$$

$$e = \langle N, x_{uu} \rangle = 0, \quad f = \langle N, x_{uv} \rangle = \frac{a}{\sqrt{1+a^2(u^2+v^2)}}, \quad g = \langle N, x_{vv} \rangle = 0$$

$$\text{so } e(0,0) = 0 = g(0,0), \quad f(0,0) = a$$

$$K = \frac{eg - f^2}{EG - F^2} = \frac{-a^2}{1 \cdot 1} = -a^2, \quad H = \frac{1}{2} \left[\frac{eG - 2fF + gE}{EG - F^2} \right] = 0$$

Q2: $x(u, v) = (v \cos(u), v \sin(u), cu)$

$$x_u = (-v \sin(u), v \cos(u), c), \quad x_v = (\cos(u), \sin(u), 0), \quad x_{uu} = (-v \cos(u), -v \sin(u), 0)$$

$$x_{vv} = (0, 0, 0), \quad x_{uv} = (-\sin(u), \cos(u), 0) = x_{vu}$$

$$E = v^2 + c^2, \quad F = 0, \quad G = 1, \quad N = \frac{1}{\sqrt{v^2 + c^2}} (-c \sin(u), c \cos(u), -v)$$

$$e = 0, \quad f = \frac{c}{\sqrt{c^2 + v^2}}, \quad g = 0$$

$$H = \frac{1}{2} \left[\frac{eg - 2fF + gE}{EG - F^2} \right] = 0, \quad K = \frac{eg - f^2}{EG - F^2} = \frac{-c^2}{(c^2 + v^2)^2} < 0$$

Thus, every point $x(u, v)$ is hyperbolic. Let $\alpha(t) = x(u(t), v(t))$, where I is a parametrisation of a connected region C . Thus, C is an asymptotic curve if the form $II(\alpha'(t)) = 0 \Leftrightarrow e(u')^2 + 2f u'v' + g(v')^2 = 0$

$$\frac{2c}{\sqrt{c^2 + v^2}} u'v' = 0 \Leftrightarrow u'v' = 0 \Leftrightarrow u = \text{constant or } v = \text{constant}$$

$$\begin{vmatrix} (v')^2 & -u'v' & (u')^2 \\ E & F & G \\ e & f & g \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} (v')^2 & -u'v' & (u')^2 \\ v^2 + c^2 & 0 & 1 \\ 0 & \frac{c}{\sqrt{c^2 + v^2}} & 0 \end{vmatrix} = 0$$

$$\frac{-c}{\sqrt{c^2 + v^2}} [(v')^2 - (u')^2 (v^2 + c^2)] = 0$$

$$-\sinh(v) \cos^2(u) - \cosh^2(v) \sin^2(u)$$

integrating b/s

$$u' = \pm \frac{v'}{\sqrt{v^2+1}} \Rightarrow u = \pm \ln(v + \sqrt{v^2+1}) + C$$

$$\Leftrightarrow u \pm \ln(v + \sqrt{v^2+1}) = \text{constant}$$

Q3:

$$X_u = (-\cosh v \sin(u), \cosh v \cos(u), 0) \quad X_v = (\sinh v \cos u, \sinh v \sin(u), 1)$$

$$X_{uv} = X_{vu} = (-\sinh v \sin(u), \sinh v \cos(u), 0)$$

$$X_{uu} = (-\cosh(v) \cos(u), -\cosh(v) \sin(u), 0)$$

$$X_{vv} = (\cosh v \cos u, \cosh v \sin(u), 0)$$

$$E = \cosh^2 v, F = 0, G = \sinh^2 v + 1, N = \frac{1}{\cosh(v)} (\cos u, \sin(u), -\sinh(v))$$

$$e=1, f=0, g=1$$

$$e(u')^2 + 2f u'v' + g(v')^2 = 0 \quad \forall t \in I$$

$$(u')^2 + (v')^2 = 0$$

$$\int (u')^2 = -\int (v')^2 \Leftrightarrow u^2 = \pm v^2 \Leftrightarrow u+v = \text{constant} \text{ or } u-v = \text{constant}$$

Q4 $X(u,v) = (u, v, uv)$

$$X_u = (1, 0, v); X_v = (0, 1, u) \quad X_{uv} = X_{vu} = (0, 0, 1) \quad X_{uu} = X_{vv} = (0, 0, 0)$$

$$E = v^2+1; F = uv; G = u^2+1; e=0, f=0, g=\frac{1}{\sqrt{u^2+v^2+1}}$$

$$N = \frac{(-v, -u, 1)}{\sqrt{u^2+v^2+1}}$$

$$e(u')^2 + 2f u'v' + g(v')^2 = 0 \quad \forall t \in I \quad \text{for a parametrization } \alpha(t) = X(u(t), v(t))$$

$$\Leftrightarrow 2u'v' = 0 \Leftrightarrow u'v' = 0 \Leftrightarrow u = \text{constant} \text{ or } v = \text{constant}$$

asymptotic curves

$$\begin{vmatrix} (v')^2 & -u'v' & (u')^2 \\ E & F & G \\ e & f & g \end{vmatrix} = 0 \Leftrightarrow \begin{vmatrix} (v')^2 & -u'v' & (u')^2 \\ v^2+1 & uv & u^2+1 \\ 0 & \frac{1}{\sqrt{1+u^2+v^2}} & 0 \end{vmatrix} = 0$$

$$\frac{1}{\sqrt{1+u^2+v^2}} \begin{vmatrix} (v')^2 & (u')^2 \\ v^2+1 & u^2+1 \end{vmatrix} = 0 \Leftrightarrow (v')^2(u^2+1) - (u')^2(v^2+1) = 0$$

$$u' = \pm \frac{1+u^2}{\sqrt{1+u^2}} v' \Rightarrow \frac{u'}{\sqrt{1+u^2}} = \pm \frac{v'}{\sqrt{1+v^2}}$$

$$\Rightarrow \operatorname{arcsinh}(u) = \pm \operatorname{arcsinh}(v) + C$$

$$\Rightarrow \operatorname{arcsinh}(u) \pm \operatorname{arcsinh}(v) = \text{constant}$$

Q5

$$X_u = (1 - u^2 + v^2, 2uv, 2u), \quad X_v = (2uv, 1 - v^2 + u^2, -2v)$$

$$X_{uu} = (-2u, 2v, 2), \quad X_{uv} = (2v, -2u, -2), \quad X_{vv} = (2v, 2u, 0)$$

$$\begin{aligned} E = \langle X_u, X_u \rangle &= (1 - u^2 + v^2)^2 + 4u^2v^2 + 4u^2 = 1 - 2u^2 + u^4 + 2(1 - v^2)(v^2) + v^4 + 4u^2v^2 \\ &= 1 - 2u^2 + u^4 + 2v^2 - 2u^2v^2 + v^4 + 4u^2v^2 + 4u^2 \\ &= 1 + 2u^2 + u^4 + 2v^2 + 2u^2v^2 + v^4 \\ &= (1 + u^2 + v^2)^2 \end{aligned}$$

$$\begin{aligned} F = 2u \langle X_u, X_v \rangle &= 2uv(1 - u^2 + v^2) + 2uv(1 + u^2 - v^2) - 4uv \\ &= 2uv(1 - u^2 + v^2 + 1 + u^2 - v^2 - 2) \\ &= 0 \end{aligned}$$

$$\begin{aligned} G = \langle X_v, X_v \rangle &= 4u^2v^2 + 1 - 2v^2 + v^4 + 2u^2 - 2u^2v^2 + u^4 + 4v^2 \\ &= 1 + 2v^2 + 2u^2 + 2u^2v^2 + v^4 + u^4 \\ &= (1 + u^2 + v^2)^2 \end{aligned}$$

$$N = \frac{X_u \wedge X_v}{|X_u \wedge X_v|} = \frac{1}{u^2 + v^2 + 1} (-2u, 2v, 1 - (u^2 + v^2))$$

$$e = \frac{4u^2 + 4v^2 + 2 - 2u^2 - 2v^2}{u^2 + v^2 + 1} = \frac{2u^2 + 2v^2 + 2}{u^2 + v^2 + 1} = 2$$

$$f = 0$$

$$g = \frac{-4u^2 - 4v^2 - 2 + 2u^2 + 2v^2}{u^2 + v^2 + 1} = \frac{-2(u^2 + v^2 + 1)}{u^2 + v^2 + 1} = -2$$

$$\begin{aligned} c) \quad k = \frac{eg - f^2}{Eg - F^2} &= \frac{-4}{(1 + u^2 + v^2)^4}; \quad h = \frac{1}{2} \frac{eG - 2fF + gE}{Eg - F^2} = \frac{1}{2} \frac{2(1 + u^2 + v^2)^2 - 2(1 + u^2 + v^2)^2}{(1 + u^2 + v^2)^4} \\ &= 0 \end{aligned}$$

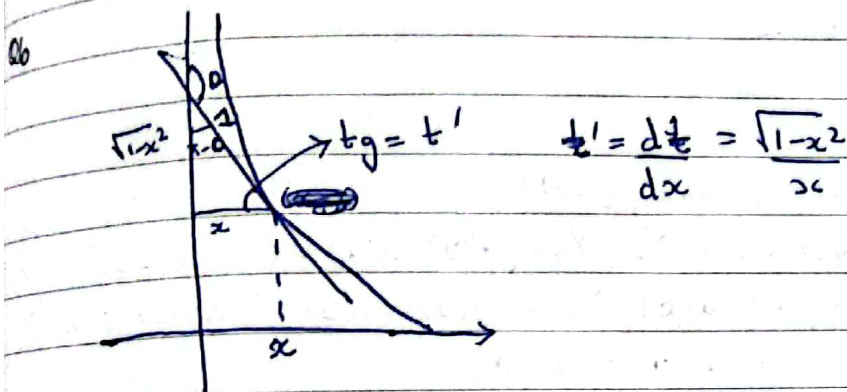
$$\begin{aligned} d) \quad x^2 - 2kx + k &= 0 \Leftrightarrow x^2 + k = 0 \Rightarrow x_1 = \sqrt{-k} \text{ and } x_2 = -\sqrt{-k} \\ \Rightarrow k_1 &= 2 \quad \& \quad k_2 = -2 \end{aligned}$$

$$d) \quad \begin{vmatrix} (uv)^2 & -u^2v & (v^2)^2 \\ (uv)^2 & 0 & (1 + u^2 + v^2)^2 \\ 2 & 0 & -2 \end{vmatrix} = 0 \Leftrightarrow uv^2 \begin{vmatrix} (1 + u^2 + v^2)^2 & (1 + u^2 + v^2)^2 \\ 2 & -2 \end{vmatrix} = 0$$

$$uv^2 = 0 \Leftrightarrow u = \text{constant or } v = \text{constant}$$

$$\begin{aligned} & \cos^2 u \quad \sin u \\ & - 2 \cos u \sin u \quad \cos u \\ & \frac{-2 \cos u + 2 \cos^3 u - \cos^3 u}{\sin^2 u} \end{aligned}$$

$$\begin{aligned} & (v'(u))^2 + 2f(u)v' + g(v')^2 = 0, \quad t \in I \\ & 2(u')^2 - 2(v')^2 = 0 \Leftrightarrow (u')^2 = (v')^2 \\ & u' = v' \text{ or } u' = -v' \text{ integrating b/s} \\ & u - v = \text{constant or } u + v = \text{constant} \end{aligned}$$



$$\begin{aligned} & \sin(\pi - \theta) = x = \sin(\theta) \\ & 1 = \cos \theta \frac{d\theta}{dx} \Rightarrow \frac{d\theta}{dx} = \frac{1}{\cos(\theta)} \end{aligned}$$

$$\text{Assume } \frac{dt}{d\theta} \cdot \frac{d\theta}{dx} = \frac{\sqrt{1-\sin^2 \theta}}{\sin \theta} \Rightarrow \frac{dt}{d\theta} = \frac{\cos^2 \theta}{\sin \theta}$$

$$\frac{dt}{d\theta} \int d\theta = \int \frac{1}{\sin \theta} d\theta - \int \sin \theta d\theta =$$

$$\begin{aligned} t &= -\ln(\csc(\theta) + \cot(\theta)) + \cos(\theta) + c \\ &= \ln\left(\tan\left(\frac{\theta}{2}\right)\right) + \cos(\theta) + c \end{aligned}$$

$$\text{at } \theta = \frac{\pi}{2}, t\left(\frac{\pi}{2}\right) = 0 \text{ hence } c = 0$$

$$\text{so } t = \ln\left(\tan\left(\frac{\theta}{2}\right)\right) + \cos(\theta)$$

$$t(\theta) = \left(\sin(\theta), \ln\left(\tan\left(\frac{\theta}{2}\right)\right) + \cos(\theta)\right)$$

$$(b) \text{ After rotating the tractrix, } X(u, v) = (\sin(u) \cos v, \sin(u) \sin v, \ln(\tan(\frac{u}{2})) \cos u)$$

It is easy to check it is regular.

$$(c) X_u = (\cos(u) \cos(v), \cos(u) \sin(v), \frac{\cos^2(u)}{\sin(u)})$$

$$X_v = (-\sin(u) \sin v, \sin(u) \cos v, 0)$$

$$X_{uv} = (-\cos(u) \sin v, \cos(u) \cos v, 0) = X_{vu}$$

$$X_{uu} = (-\sin(u) \cos(v), -\sin(u) \sin(v), \frac{-\cos^3(u)}{\sin^2(u)} - 2 \cos(u))$$

$$X_{vv} = (-\sin(u) \cos v, -\sin(u) \sin v, 0)$$

$$E = \frac{\cos^2 u}{\sin^2 u}, F = 0, G = \sin^2 u$$

$$e = \frac{-\cos(u)}{\sin(u)}, f = 0, g = \sin(u) \cos(u)$$

$$K = \frac{eg - f^2}{EG - F^2} = \frac{-\cos^2 u}{\cos^2 u} = -1$$

(a) Q7: $X(u, v) = (\varphi(v) \cos(u), \varphi(v) \sin(u), \psi(v))$

$$X_u = (-\varphi(v) \sin(u), \varphi(v) \cos(u), 0), X_v = (\varphi'(v) \cos(u), \varphi'(v) \sin(u), \psi'(v))$$

$$X_{uu} = (-\varphi(v) \cos(u), -\varphi(v) \sin(u), 0), X_{uv} = (\varphi''(v) \cos(u), \varphi''(v) \sin(u), \varphi'(v))$$

$$X_{vu} = X_{uv} = (-\varphi'(v) \sin(u), \varphi'(v) \cos(u), 0)$$

$$E = \varphi(v)^2, F = 0, G = (\varphi'(v))^2 + (\psi'(v))^2 = 1$$

$$e = -\varphi(v) \psi'(v), f = 0, g = \varphi''(v) \varphi'(v) - \psi''(v) \psi'(v)$$

$$K = \frac{eg - f^2}{EG - F^2} = \frac{-\varphi(v) \psi'(v) (\varphi''(v) \varphi'(v) - \psi''(v) \psi'(v))}{\varphi(v)^2}$$

$$= -\psi'(v) (\varphi''(v) \varphi'(v) - \psi''(v) \psi'(v))$$

$$= -\psi'(v) (\varphi''(v) \varphi'(v) - \psi''(v) \psi'(v))$$

$$\varphi(v)$$

integrating $(\varphi')^2 + (\psi')^2 = 1$ gives us $2\varphi'\varphi'' + 2\psi'\psi'' = 0$

$$\varphi'\varphi'' = -\psi'\psi''$$

$$K = \frac{-(\varphi')^2 \psi'' + \psi' \psi'' (\varphi')^2}{\varphi(v)}$$

$$= -(\varphi')^2 \psi'' - (\psi')^2 \varphi''$$

$$K = \frac{-\psi''((\varphi')^2 + (\psi')^2)}{\varphi} = -\frac{\psi''}{\varphi}$$

$$\varphi K + \psi'' = 0$$

$$(\varphi')^2 + (\psi')^2 = 1 \Rightarrow (\psi')^2 = 1 - (\varphi')^2 \Rightarrow \psi = \int \sqrt{1 - (\varphi')^2} dv$$

$$\varphi'' + \varphi K = 0 \xrightarrow{K=0} \varphi'' + \varphi = 0 \text{ so } \varphi = A \cos(v) + B \sin(v)$$

$$\psi'(v) = B \cos(v) - A \sin(v)$$

$$\psi'(0) = B$$

$$\langle \varphi'(0), \psi'(0) \rangle = 0 \Leftrightarrow \langle \varphi'(0), \psi'(0) \rangle, (1, 0) = 0$$

$$\psi(v) = A \cos(v) \text{ so } C = \psi(0) \quad \{ B = 0$$

$$\psi(v) = \int \sqrt{1 - C^2 \sin^2 v} \, dv \quad 1 - C^2 \sin^2 v \geq 0$$

Substituting
A with C

$$C^2 \sin^2 v \leq 1 \Rightarrow \sin^2 v \leq \frac{1}{C^2}$$

Since that's what they used. $\frac{1}{|C|} \leq \sin(v) \leq \frac{1}{|C|}$

$$\Rightarrow \sin^{-1}\left(\frac{1}{|C|}\right) \leq v \leq \sin^{-1}\left(\frac{1}{|C|}\right) \Rightarrow C \neq 0$$

(c) $K = -1$

$$\psi'' - \psi = 0 \Rightarrow \psi = C_1 e^v + C_2 e^{-v}$$

$$C_1 = C_2 = \frac{C}{2} \quad \psi(v) = \frac{C}{2} e^v + e^{-v} = \frac{C}{2} \cosh v = C \cosh(v) \Rightarrow \psi(v) = \int_0^v \sqrt{1 - C^2 \sinh^2 t} \, dt$$

$$C_1 = -C_2 = \frac{C}{2}$$

$$\psi(v) = C \left(\frac{e^v - e^{-v}}{2} \right) = C \sinh(v) \Rightarrow \psi(v) = \int_0^v \sqrt{1 - C^2 \cosh^2 t} \, dt$$

$$C_1 = 1, C_2 = 0 \Rightarrow \psi(v) = e^v \Rightarrow \psi = \int_0^v \sqrt{1 - e^{2v}} \, dv$$